HEIQ XMEMBRANE PROJECT TEAM

EO vapour experiment design

Moisture target, coupon-cell scaling and reversible cycling

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Experiment context

This report translates the selected O30 building-scale ERV+EO result into a measurable coupon experiment. EO is located inside the ERV membrane/core. The report explains the moisture-rate target, active-area selection, channel and fixture dimensions, detectability basis, transient closed-cell response and the limitations of symmetric water shuttling without through-flow.

Primary decision. Build one modular fixture around a 50 × 50 mm maximum active aperture (25 cm^2). Use controlled once-through gas to determine sustained polarity-dependent EO transport. Use a sealed or recirculating configuration only as a separate transient experiment: symmetric polarity cycling moves water in both directions but gives zero net EO transport over a complete cycle unless a rectifying or water-removal mechanism is added.

System-derived net EO target	200–250 g/m^2/h below 1 V
Maximum active aperture	25 cm^2; physical coupon approximately 80 × 80 mm
Nominal gas state	32 °C, 101,325 Pa, water activity 0.50
Nominal channel	2 mm gap per side; 1/2/4-mm validation sweep
Nominal gas flow	6 L/min per side for the declared detectability screen
Claim status	Experiment-design target; no measured membrane or commercial-system validation

Executive summary

The O30 system coordinate requires 4,365.9 kg/h total water transfer through the active ERV membrane. The matched passive ERV provides approximately 3,137.6 kg/h, leaving 1,228.3 kg/h as the additional duty associated with EO. Dividing that incremental duty by the selected 5,899 m^2 active area gives 208 $g/m^2/h$. This is a system-equivalent average, not an intrinsic membrane constant or a uniform local flux.

For the 25-cm^2 coupon, 208 $g/m^2/h$ equals 0.520 g/h or 0.144 mg/s. The ideal gross mapping at transport number $t_w = 50$ and current density $j = 10$ A/m^2 gives 336 $g/m^2/h$, corresponding to 0.840 g/h or 0.233 mg/s on the same coupon. Gross flux is not net flux: passive back-diffusion, gas films, storage, leakage and electrode chemistry must be resolved experimentally.

The 25-cm^2 aperture is retained because it gives the largest absolute signal and lowest perimeter-to-area ratio in the existing fixture. At 6 L/min per side, the ideal inverse-design anchor needs 13.09 cm^2 at 0.5 A/m^2 and 6.55 cm^2 at 1 A/m^2 to exceed the declared five-to-one humidity-signal criterion. Order-unity transport numbers require lower flow or an integrating water-rate measurement rather than a larger permanent fixture.

208 $g/m^2/h$
System-equivalent EO increment
Use 200–250 as the coupon target

0.520 g/h
25-cm^2 target rate
0.144 mg/s at 208 $g/m^2/h$

0 net/cycle
Symmetric sealed cycling
Water shuttles, then returns

Interpretation boundary. A closed-cell oscillation can demonstrate directionality, reversibility and finite water displacement. It cannot by itself demonstrate sustained ERV dehumidification. The internal gas inventory is small, the humidity gradient develops rapidly, blocking electrodes charge, and a symmetric full cycle returns the EO-displaced water.

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1 From building moisture duty to a coupon target

For dry-air mass flow $\dot{m}_{da}$ and humidity ratios w, the required water rate is

$$\dot{m}_w = \dot{m}_{da}(w_{\text{in}} - w_{\text{target}})3600.$$

Table 1: O30 moisture balance used to define the experimental target.

Quantity	Value	Meaning
Total active-ERV membrane transport	4,365.9 kg/h	Passive plus electrically enhanced transfer
Matched passive-ERV transport	≈3,137.6 kg/h	Same ERV with EO switched off
Additional duty associated with EO	1,228.3 kg/h	Difference between total and passive cases
Selected active membrane area	5,899 m^2	Coupled system-sweep result
Average total membrane flux	740 g/m^2/h	Total transport divided by area
Area-normalised EO increment	208 g/m^2/h	Additional EO duty divided by area

$$J_{\text{total}} = \frac{4{,}365.9}{5{,}899} = 0.740\ \text{kg}\,\text{m}^{-2}\,\text{h}^{-1}, \qquad J_{\text{EO,eq}} = \frac{1{,}228.3}{5{,}899} = 0.208\ \text{kg}\,\text{m}^{-2}\,\text{h}^{-1}.$$

The 5,899-m^2 area is generated by a coupled ERV model with axially varying activity, temperature, current, conductivity, gas films and back-diffusion. It is not obtained by dividing the water duty by a single measured membrane flux. Consequently, the experimental requirement is stated as a band: demonstrate 200–250 g/m^2/h of repeatable net polarity-odd transport below 1 V under representative gas–gas conditions.

2 Transport number, current and sensitivity

The ideal gross EO relationship is

$$J_{\text{EO,gross}} = t_w \frac{j}{F} M_w \approx 0.672\, t_w j \quad \text{g}\,\text{m}^{-2}\,\text{h}^{-1},$$

where t_w is mol water per mol elementary charge and j is in A/m^2. This is a gross Faradaic mapping; it does not subtract passive diffusion, films, leakage, water storage or reaction water.

Table 2: Gross EO sensitivity at 10 A/m^2.

t_w	Gross EO flux [g/m^2/h]
10	67
20	134
30	202
50	336
75	504
100	672

The ideal 208-g/m^2/h target corresponds to $t_w j \approx 310$. Examples are $t_w = 50$ at 6.2 A/m^2, $t_w = 30$ at 10.3 A/m^2, and $t_w = 20$ at 15.5 A/m^2. A real experiment can require a higher product because the observable net flux is the gross EO contribution minus opposing transport.

First-order sensitivities. Gross flux scales linearly with t_w and j. Total rate scales linearly with active area at unchanged local conditions. Voltage, heating, concentration polarisation, conductivity and back-diffusion need not scale linearly. A high liquid-state t_w is therefore a screening lead, not a validated vapour-state coefficient.

3 Cell area and measurable water rate

For uniform flux J and active area A, $\dot{m} = JA$. Table 3 compares the system-equivalent target with the ideal $t_w = 50$, $j = 10$ gross reference.

Table 3: Coupon rate as a function of active aperture.

Aperture	Area	Rate at 208 g/m^2/h	Rate at 336 g/m^2/h
20 × 20 mm	4 cm^2	0.083 g/h; 0.023 mg/s	0.134 g/h; 0.037 mg/s
30 × 30 mm	9 cm^2	0.187 g/h; 0.052 mg/s	0.302 g/h; 0.084 mg/s
40 × 40 mm	16 cm^2	0.333 g/h; 0.092 mg/s	0.538 g/h; 0.149 mg/s
50 × 50 mm	25 cm^2	0.520 g/h; 0.144 mg/s	0.840 g/h; 0.233 mg/s

The aperture sweep should be fitted to

$$\dot{m}_{\mathrm{measured}} = J_A A + k_P P + \dot{m}_{\mathrm{blank}},$$

where P is perimeter. The 4-cm^2 square and 8 × 50-mm rectangle have equal area but different perimeter and flow orientation. They help distinguish membrane-area transport from seal, edge and distribution artefacts.

4 Detectability basis for the 25-cm^2 fixture

The declared sizing state is 32 °C, 101,325 Pa, inlet water activity 0.50 and 6 L/min per side. Each humidity channel has 0.1 RH percentage point expanded uncertainty. For two independent channels,

$$u_{\Delta RH} = \sqrt{0.1^2 + 0.1^2} = 0.141, \qquad \Delta RH_{\mathrm{required}} = 5u_{\Delta RH} = 0.707 \text{ percentage point.}$$

At fixed flow, the ideal outlet signal scales approximately as $t_w j A/\dot{V}$. The sizing calculation converts gross water rate to humidity ratio and RH at the declared state, then solves for the area that reaches the 0.707-point threshold.

Table 4: Minimum ideal active area at 6 L/min per side.

Gross-signal case	Minimum area	Smallest insert
Inverse-design anchor, 0.5 A/m^2	13.09 cm^2	16 cm^2
Inverse-design anchor, 1.0 A/m^2	6.55 cm^2	9 cm^2
Selected ERV profile minimum, 1.242 A/m^2	5.27 cm^2	9 cm^2
Selected ERV profile maximum, 3.366 A/m^2	1.94 cm^2	4 cm^2
Order-unity $t_w = 1$, 10 A/m^2	65.5 cm^2	Does not fit
Low-drag $t_w = 0.3$, 10 A/m^2	218 cm^2	Does not fit

These are ideal signal floors, not net-flux predictions. With the 25-cm^2 aperture, the maximum ideal flows are approximately 2.29 L/min for $t_w = 1$ and 0.69 L/min for $t_w = 0.3$ at 10 A/m^2. Low-flow signal acquisition and high-flow film characterisation should be separate modes.

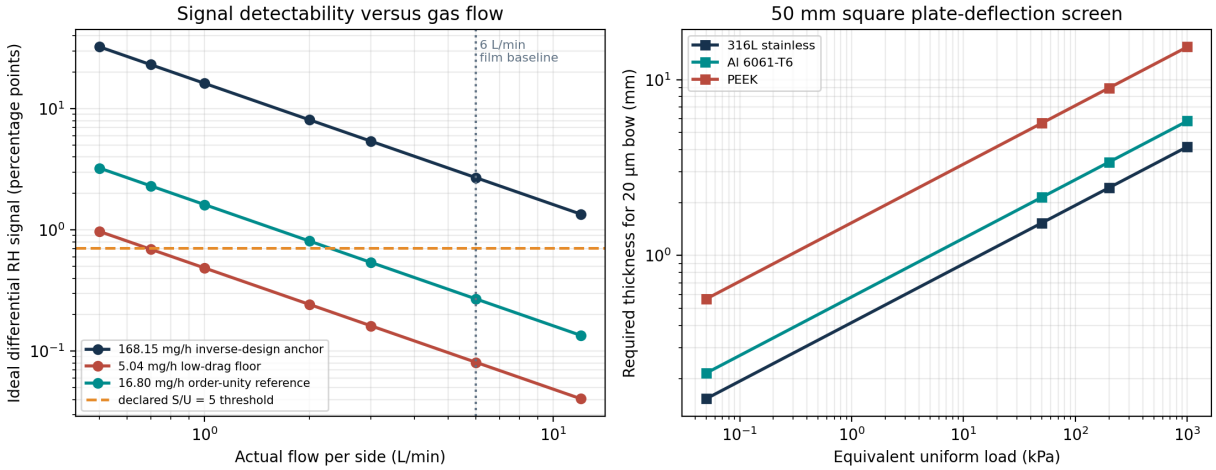


Figure 1: Existing project calculation of cell dimensions and measurement trade-offs. Values are experiment-design sensitivities, not validated material performance.

5 Mechanical and fluidic starting point

Table 5: Fixture dimensions and their calculation basis.

Item	Starting value	Basis
Maximum active aperture	50 × 50 mm	Signal margin; low perimeter/area ratio
Physical membrane coupon	≈80 × 80 mm	Double seals, guard groove and clamp land
Outer fixture	≈110 × 110 mm	Ports, bolts and electrical isolation
Nominal gap per side	2 mm	Local match to declared ERV channel
Gap validation	1, 2 and 4 mm	Gas-film and pressure-drop sensitivity
Active volume per side	5.00 mL	25 cm^2 × 0.2 cm
Headers, ports and taps	1.222 mL	Proposed geometric allowance
Total swept volume	6.222 mL/side	Active channel plus internal dead volume
Ideal residence at 6 L/min	62 ms	6.222 mL / 100 mL/s
Main pressure difference	within ±5 Pa	Limit hydraulic transport artefacts

Hard stops, not gasket compression, must define the gas gap. For the 1-mm gap, the plate-bow target is below 20 μm or the measured nonuniformity must be included in analysis. The stiffness screen gives 15.35 mm PEEK, 5.81 mm Al 6061-T6 and 4.16 mm 316L at a deliberately conservative 1-MPa equivalent uniform load. These are screening values, not gasket-contact calculations; final acceptance requires measured flatness under the real assembly load.

6 Closed-cell fluctuation without through-flow

At constant flux, the water shifted during one half-cycle is

$$\Delta m_{\mathrm{half}} = JA\Delta t = \frac{JA}{2f},$$

where f is the full-cycle frequency. For 25 cm^2 at the 208-g/m^2/h target, $JA = 0.520$ g/h = 0.144 mg/s.

Multiply these values by 1.615 for the 336-g/m^2/h gross reference. These linear values apply only while current and boundary states remain available. They are initial command-rate calculations, not guaranteed cycle amplitudes.

The 6.222-mL internal gas volume contains only about 0.1–0.2 mg water vapour over the full 0–100% RH span at 25–32 °C. At 0.144 mg/s, the chamber humidity can therefore change materially in roughly one second without an external reservoir,

Table 6: Ideal one-direction displacement for the 25-cm^2 coupon.

Frequency	Half-cycle	Water shifted
0.1 Hz	5 s	0.72 mg
0.01 Hz	50 s	7.2 mg
0.001 Hz	500 s	72 mg
0.0001 Hz	5,000 s	722 mg

sorbent or condensate capacity. The resulting activity difference drives back-diffusion. Tubing, sensors, electrodes, supports and walls must be included in the transient water inventory.

Full-cycle result. In the current constant-drag model, symmetric charge-balanced polarity cycling gives

$$\boxed{\text{net EO water transport over one complete cycle} = 0.}$$

Absolute shuttled water is nonzero, but the second half-cycle returns the EO displacement. Continuous dehumidification requires a validated mechanism that breaks this symmetry, such as flow switching, condensation, rectification, asymmetric regeneration, pseudocapacitance or a defined Faradaic pathway.

7 Blocking-electrode capacity

For purely blocking electrodes, the ideal finite displacement is

$$m_{\max} = t_w \frac{M_w}{F} C_{\text{pair},A} A \Delta V.$$

The reference sensitivity uses 10,000 F/m^2 for each identical electrode, or 5,000 F/m^2 for the series pair. For 25 cm^2, $t_w \approx 100$ and a 1.8-V terminal limit, the transient model gives about 420 mg EO displacement before current collapses. The same ideal scaling at $t_w = 50$ gives approximately 210 mg. These capacitances and transport numbers are unmeasured sensitivity coordinates.

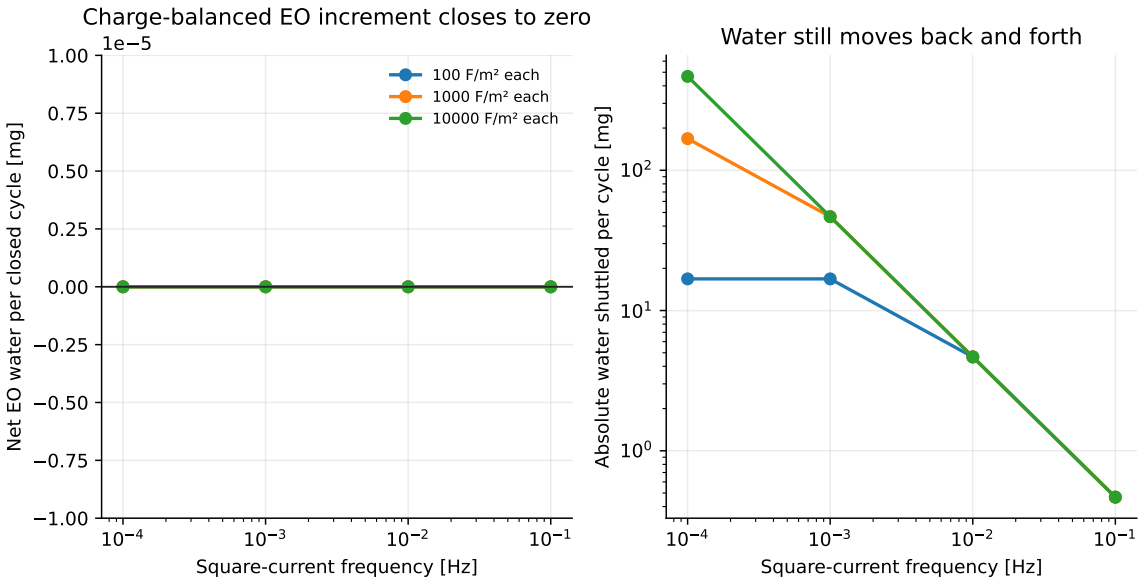


Figure 2: Reference symmetric cycle from the existing transient study. Charge-balanced reversal closes the EO water balance over a full cycle.

8 Sensitivity map and experimental meaning

Table 7: First-order sensitivity and validity limits.

Variable increased	First-order effect	Principal limitation
Active area	Rate and capacitive inventory rise linearly	Edge leakage and current uniformity
Transport number	Gross flux per charge rises linearly	Must be measured in vapour
Current density	Initial gross flux rises linearly	Voltage, heating, polarisation, reactions
Half-cycle time	Ideal displacement rises linearly	Chamber/electrode saturation
Frequency	Displacement per half-cycle falls as $1/f$	Fast cycling may probe electrodes, not membrane
Pair capacitance	Blocking-electrode inventory rises linearly	Still zero net in a symmetric full cycle
Gas flow	RH signal decreases at fixed water rate	Films and boundary control may improve
Channel gap	Changes volume, velocity, films and pressure drop	Must be measured in the 1/2/4-mm sweep
Diffusivity	Faster redistribution and passive flux	Back-diffusion can reduce net EO signal

The membrane redistribution scale is approximately L^2/D. For the 36.9-μm reference membrane, the present diffusivity screen spans seconds to roughly one day. Steady state must therefore be defined by stable voltage and closed donor/receiver water balances, not by one fixed elapsed time.

9 Recommended test programme

1. Run impermeable-foil, zero-current and passive-gradient blanks.
2. Use once-through gas at equal inlet activity and low current to locate a measurable signal safely.
3. Apply both polarities in an ABBA sequence and calculate $J_{\mathrm{odd}} = [J(+I) - J(-I)]/2$.
4. Require donor loss and receiver gain to close independently.
5. Repeat the 4/9/16/25-cm^2 aperture sweep and fit area, perimeter and blank terms.
6. Sweep flow separately from the 1/2/4-mm gap sweep to expose gas-film limitations.
7. After sustained polarity-dependent transport is established, connect two measured, well-mixed reservoirs and test reversible half-cycles.
8. Report initial rate, half-cycle displacement, peak-to-peak inventory, full-cycle water closure, charge closure, voltage and energy.

Proposed first success criterion. Demonstrate repeatable polarity-odd net transport of at least 200–250 g/m^2/h below 1 V under representative gas–gas conditions; show matching donor loss and receiver gain; and show that the rate is dominated by active area rather than perimeter or blank terms.

The resulting report should also give effective $t_w = FJ_{\mathrm{odd}}/(jM_w)$, conductivity or area-specific resistance, passive water flux, temperature, activity, pressure, flow, gap, electrode state and measurement uncertainty for the same specimen.

10 Conclusions and claim boundary

The system model supplies a clear experimental scale: approximately 208 g/m^2/h additional EO-equivalent transport at O30, rounded to a practical 200–250-g/m^2/h target. The 25-cm^2 coupon converts this into a measurable target rate of 0.520 g/h while retaining smaller apertures for area/perimeter validation.

The fixture dimensions are driven by signal, boundary control, seal land, channel similarity and artefact separation. They do not validate the assumed high transport number. The decisive evidence must be a vapour-state, polarity-dependent donor/receiver water balance under measured voltage and current.

Closed-cell oscillation is useful but answers a different question. It measures finite displacement, reversal and storage. Without through-flow or another asymmetry, it cannot provide sustained net dehumidification. Absolute water shuttled per half-cycle must never be reported as continuous moisture-removal capacity.

> All numerical material coordinates in this report remain conditional until measured on the same membrane under representative vapour activity, temperature, current density and interface conditions. The report defines an experiment and falsifiable targets; it does not claim a validated membrane, electrode, receiver or commercial ERV system.

Reproduction and source documents

The calculations are reproduced with:

```
python3 scripts/design_eo_vapor_test_cell.py
PYTHONPATH=src python3 scripts/run_capacitive_electrode_study_v1.py --overwrite
pytest -q -s tests/test_eo_vapor_test_cell_design.py
pytest -q -s tests/test_capacitive_electrode_study_v1.py
```

The detailed bases are:

- `docs/EO_VAPOR_TEST_CELL_DESIGN.md`;
- `docs/TEST_CELL_AND_ERV_SCALE_CHECK_2026-09-01.md`;
- `docs/CAPACITIVE_ELECTRODE_TRANSIENT_STUDY_2026-09-01.md`; and
- `docs/030_REQUIRED_AREA_1V_v1.md`.